

sg13g2_io_slow_1p35V_3p0V_125C Library

Cell Groups
INOUTx
INPUT
SG13G2_CORNER
SG13G2_FILLER10000
SG13G2_FILLER1000
SG13G2_FILLER2000
SG13G2_FILLER200
SG13G2_FILLER4000
SG13G2_FILLER400
SG13G2_IOPADANALOG
SG13G2_IOPADIOVDD
SG13G2_IOPADIOVSS
SG13G2_IOPADVDD
SG13G2_IOPADVSS
TRI_OUTx

INOUTx



*sg13g2_io_slow_1p35V_3p0V_125C Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C, Voltage 1.35, Temp
125.00*

Truth Table

INPUT			OUTPUT	
c2p	c2p_en	pad	pad	p2c
-	0	0	-	0
-	0	1	-	1
0	1	-	0	0
1	1	-	1	1

Footprint

Cell Name	Area
sg13g2_IOPadInOut16mA	14400.00000
sg13g2_IOPadInOut30mA	14400.00000
sg13g2_IOPadInOut4mA	14400.00000
sg13g2_IOPadOut16mA	14400.00000
sg13g2_IOPadOut30mA	14400.00000
sg13g2_IOPadOut4mA	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)			Max Cap(pf)	
	c2p	c2p_en	pad	p2c	pad
sg13g2_IOPadInOut16mA	0.02666	0.02564	0.30434	0.24000	20.00000
sg13g2_IOPadInOut30mA	0.02666	0.02563	0.39755	0.24000	40.00000
sg13g2_IOPadInOut4mA	0.02666	0.02564	0.22409	0.24000	12.00000
sg13g2_IOPadOut16mA	0.03887	0.00000	0.00000	0.00000	20.00000
sg13g2_IOPadOut30mA	0.03889	0.00000	0.00000	0.00000	40.00000
sg13g2_IOPadOut4mA	0.03888	0.00000	0.00000	0.00000	12.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadInOut16mA	707.77300	8739.08972	17609.50000
sg13g2_IOPadInOut30mA	787.83300	9799.98817	17609.60000
sg13g2_IOPadInOut4mA	726.62600	7875.97533	17609.50000
sg13g2_IOPadOut16mA	1094.83000	3646.92833	6176.24000
sg13g2_IOPadOut30mA	1533.64000	4783.27000	9084.32000
sg13g2_IOPadOut4mA	800.31500	2712.52417	6176.06000

Delay Information

Delay(ns) to p2c rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad->p2c (RR)	0.12000	0.02400	0.29465	0.80000	0.14400	0.51842	5.00000	0.24000	0.92974
sg13g2_IOPadInOut30mA	pad->p2c (RR)	0.12000	0.02400	0.29477	0.80000	0.14400	0.51787	5.00000	0.24000	0.92987
sg13g2_IOPadInOut4mA	pad->p2c (RR)	0.12000	0.02400	0.29515	0.80000	0.14400	0.51768	5.00000	0.24000	0.92963

Delay(ns) to p2c falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad->p2c (FF)	0.12000	0.02400	0.30139	0.80000	0.14400	0.50895	5.00000	0.24000	0.84430
sg13g2_IOPadInOut30mA	pad->p2c (FF)	0.12000	0.02400	0.30135	0.80000	0.14400	0.50914	5.00000	0.24000	0.84450
sg13g2_IOPadInOut4mA	pad->p2c (FF)	0.12000	0.02400	0.30130	0.80000	0.14400	0.50867	5.00000	0.24000	0.84408

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p->pad (RR)	0.02000	1.00000	2.24583	0.33000	9.00000	3.77401	2.50000	20.00000	5.65392
	c2p_en->pad (FR)	0.02000	1.00000	1.59826	0.33000	9.00000	1.66072	2.50000	20.00000	1.84328
	c2p_en->pad (RR)	0.02000	1.00000	2.23533	0.33000	9.00000	3.79779	2.50000	20.00000	5.73584
sg13g2_IOPadInOut30mA	c2p->pad (RR)	0.02000	2.00000	2.73413	0.33000	18.00000	4.57538	2.50000	40.00000	6.60657
	c2p_en->pad (FR)	0.02000	2.00000	1.85519	0.33000	18.00000	1.91923	2.50000	40.00000	2.10377
	c2p_en->pad (RR)	0.02000	2.00000	2.71132	0.33000	18.00000	4.59375	2.50000	40.00000	6.65332
sg13g2_IOPadInOut4mA	c2p->pad (RR)	0.02000	1.00000	2.31966	0.33000	5.00000	5.02655	2.50000	12.00000	9.72590
	c2p_en->pad (FR)	0.02000	1.00000	1.33209	0.33000	5.00000	1.39044	2.50000	12.00000	1.58170
	c2p_en->pad (RR)	0.02000	1.00000	2.32690	0.33000	5.00000	5.10699	2.50000	12.00000	9.94536
sg13g2_IOPadOut16mA	c2p->pad (RR)	0.02000	1.00000	2.17211	0.33000	9.00000	3.77877	2.50000	20.00000	6.44789
sg13g2_IOPadOut30mA	c2p->pad (RR)	0.02000	2.00000	2.66612	0.33000	18.00000	4.58243	2.50000	40.00000	7.40393
sg13g2_IOPadOut4mA	c2p->pad (RR)	0.02000	1.00000	2.23193	0.33000	5.00000	5.01610	2.50000	12.00000	10.50560

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p->pad (FF)	0.02000	1.00000	1.65775	0.33000	9.00000	3.04027	2.50000	20.00000	5.12155
	c2p_en->pad (FF)	0.02000	1.00000	1.60257	0.33000	9.00000	1.68885	2.50000	20.00000	1.94460
	c2p_en->pad (RF)	0.02000	1.00000	1.40977	0.33000	9.00000	2.89141	2.50000	20.00000	4.76950
sg13g2_IOPadInOut30mA	c2p->pad (FF)	0.02000	2.00000	2.20823	0.33000	18.00000	3.75553	2.50000	40.00000	5.90307
	c2p_en->pad (FF)	0.02000	2.00000	2.48017	0.33000	18.00000	2.52996	2.50000	40.00000	2.81812
	c2p_en->pad (RF)	0.02000	2.00000	1.67766	0.33000	18.00000	3.37453	2.50000	40.00000	5.37676
sg13g2_IOPadInOut4mA	c2p->pad (FF)	0.02000	1.00000	1.78320	0.33000	5.00000	4.49137	2.50000	12.00000	9.42341
	c2p_en->pad (FF)	0.02000	1.00000	0.87353	0.33000	5.00000	0.95422	2.50000	12.00000	1.22911
	c2p_en->pad (RF)	0.02000	1.00000	1.78016	0.33000	5.00000	4.55875	2.50000	12.00000	9.37118
sg13g2_IOPadOut16mA	c2p->pad (FF)	0.02000	1.00000	1.57096	0.33000	9.00000	2.92485	2.50000	20.00000	4.85150
sg13g2_IOPadOut30mA	c2p->pad (FF)	0.02000	2.00000	2.12478	0.33000	18.00000	3.65093	2.50000	40.00000	5.63502
sg13g2_IOPadOut4mA	c2p->pad (FF)	0.02000	1.00000	1.66432	0.33000	5.00000	4.35328	2.50000	12.00000	9.12463

Power Information

Internal switching power(pJ) to p2c rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad	0.12000	0.02400	-1.12603	0.80000	0.14400	-1.08122	5.00000	0.24000	-0.69520
	pad	0.12000	0.02400	0.02940	0.80000	0.14400	0.03067	5.00000	0.24000	0.03059
sg13g2_IOPadInOut30mA	pad	0.12000	0.02400	-1.76185	0.80000	0.14400	-1.71222	5.00000	0.24000	-1.32587
	pad	0.12000	0.02400	0.02969	0.80000	0.14400	0.03094	5.00000	0.24000	0.03077
sg13g2_IOPadInOut4mA	pad	0.12000	0.02400	-0.61156	0.80000	0.14400	-0.56748	5.00000	0.24000	-0.18359
	pad	0.12000	0.02400	0.02936	0.80000	0.14400	0.03019	5.00000	0.24000	0.03034

Internal switching power(pJ) to p2c falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	pad	0.12000	0.02400	1.49413	0.80000	0.14400	1.53599	5.00000	0.24000	1.91480
	pad	0.12000	0.02400	0.07339	0.80000	0.14400	0.07758	5.00000	0.24000	0.09851
sg13g2_IOPadInOut30mA	pad	0.12000	0.02400	2.12644	0.80000	0.14400	2.16608	5.00000	0.24000	2.54855
	pad	0.12000	0.02400	0.07321	0.80000	0.14400	0.07732	5.00000	0.24000	0.09824
sg13g2_IOPadInOut4mA	pad	0.12000	0.02400	0.98075	0.80000	0.14400	1.02442	5.00000	0.24000	1.39997
	pad	0.12000	0.02400	0.07349	0.80000	0.14400	0.07776	5.00000	0.24000	0.09864

Internal switching power(pJ) to pad rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p	0.02000	1.00000	5.27597	0.33000	9.00000	5.15565	2.50000	20.00000	4.92411
	c2p	0.02000	1.00000	1.89301	0.33000	9.00000	16.28530	2.50000	20.00000	30.15810
	c2p_en	0.02000	1.00000	4.86827	0.33000	9.00000	5.54010	2.50000	20.00000	7.22589
	c2p_en	0.02000	1.00000	1.85601	0.33000	9.00000	16.26200	2.50000	20.00000	30.17550
sg13g2_IOPadInOut30mA	c2p	0.02000	2.00000	8.02507	0.33000	18.00000	7.54518	2.50000	40.00000	6.90732
	c2p	0.02000	2.00000	3.69575	0.33000	18.00000	31.51310	2.50000	40.00000	14.23860
	c2p_en	0.02000	2.00000	7.24610	0.33000	18.00000	8.11320	2.50000	40.00000	8.24592
	c2p_en	0.02000	2.00000	3.65964	0.33000	18.00000	31.49240	2.50000	40.00000	14.22560
sg13g2_IOPadInOut4mA	c2p	0.02000	1.00000	3.07455	0.33000	5.00000	3.30633	2.50000	12.00000	3.74832
	c2p	0.02000	1.00000	1.89734	0.33000	5.00000	9.10611	2.50000	12.00000	21.88960
	c2p_en	0.02000	1.00000	2.85163	0.33000	5.00000	3.60699	2.50000	12.00000	5.39874
	c2p_en	0.02000	1.00000	1.86123	0.33000	5.00000	9.07240	2.50000	12.00000	21.99020
sg13g2_IOPadOut16mA	c2p	0.02000	1.00000	4.94358	0.33000	9.00000	4.80195	2.50000	20.00000	4.64931
	c2p	0.02000	1.00000	-0.02016	0.33000	9.00000	-0.01002	2.50000	20.00000	0.13424
sg13g2_IOPadOut30mA	c2p	0.02000	2.00000	7.78521	0.33000	18.00000	7.61256	2.50000	40.00000	7.03950
	c2p	0.02000	2.00000	-0.02016	0.33000	18.00000	-0.01002	2.50000	40.00000	0.13310
sg13g2_IOPadOut4mA	c2p	0.02000	1.00000	2.60912	0.33000	5.00000	2.51301	2.50000	12.00000	2.48820
	c2p	0.02000	1.00000	-0.02016	0.33000	5.00000	-0.00962	2.50000	12.00000	0.13335

Internal switching power(pJ) to pad falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadInOut16mA	c2p	0.02000	1.00000	11.24310	0.33000	9.00000	7.55593	2.50000	20.00000	6.51853
	c2p	0.02000	1.00000	0.22769	0.33000	9.00000	0.24371	2.50000	20.00000	0.39970
	c2p_en	0.02000	1.00000	2.91114	0.33000	9.00000	3.09096	2.50000	20.00000	3.38177
	c2p_en	0.02000	1.00000	0.12161	0.33000	9.00000	0.13880	2.50000	20.00000	0.23569
sg13g2_IOPadInOut30mA	c2p	0.02000	2.00000	40.03380	0.33000	18.00000	24.46550	2.50000	40.00000	18.40310
	c2p	0.02000	2.00000	0.22952	0.33000	18.00000	0.24444	2.50000	40.00000	0.38988
	c2p_en	0.02000	2.00000	4.39915	0.33000	18.00000	4.59171	2.50000	40.00000	4.89371
	c2p_en	0.02000	2.00000	0.12236	0.33000	18.00000	0.13900	2.50000	40.00000	0.22573
sg13g2_IOPadInOut4mA	c2p	0.02000	1.00000	2.25819	0.33000	5.00000	2.66728	2.50000	12.00000	3.41698
	c2p	0.02000	1.00000	0.23268	0.33000	5.00000	0.25912	2.50000	12.00000	0.42720
	c2p_en	0.02000	1.00000	1.74782	0.33000	5.00000	2.16602	2.50000	12.00000	2.91195
	c2p_en	0.02000	1.00000	0.12766	0.33000	5.00000	0.15303	2.50000	12.00000	0.26103
sg13g2_IOPadOut16mA	c2p	0.02000	1.00000	11.72830	0.33000	9.00000	7.29715	2.50000	20.00000	5.96180
	c2p	0.02000	1.00000	0.07593	0.33000	9.00000	0.08982	2.50000	20.00000	0.23376
sg13g2_IOPadOut30mA	c2p	0.02000	2.00000	41.36780	0.33000	18.00000	24.59110	2.50000	40.00000	18.28580
	c2p	0.02000	2.00000	0.07592	0.33000	18.00000	0.08982	2.50000	40.00000	0.23258
sg13g2_IOPadOut4mA	c2p	0.02000	1.00000	1.92630	0.33000	5.00000	1.87268	2.50000	12.00000	1.85699
	c2p	0.02000	1.00000	0.07596	0.33000	5.00000	0.08984	2.50000	12.00000	0.23179

Passive power(pJ) for c2p rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	-0.00009	0.33000	-0.00007	2.50000	-0.00007
	0.02000	-0.02251	0.33000	-0.02360	2.50000	-0.02387
sg13g2_IOPadInOut30mA	0.02000	-0.00010	0.33000	-0.00007	2.50000	-0.00007
	0.02000	-0.02251	0.33000	-0.02361	2.50000	-0.02387
sg13g2_IOPadInOut4mA	0.02000	-0.00009	0.33000	-0.00007	2.50000	-0.00007
	0.02000	-0.02243	0.33000	-0.02360	2.50000	-0.02387

Passive power(pJ) for c2p falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	0.00009	0.33000	0.00007	2.50000	0.00007
	0.02000	0.02737	0.33000	0.02676	2.50000	0.02671
sg13g2_IOPadInOut30mA	0.02000	0.00010	0.33000	0.00007	2.50000	0.00007
	0.02000	0.02737	0.33000	0.02676	2.50000	0.02671
sg13g2_IOPadInOut4mA	0.02000	0.00009	0.33000	0.00007	2.50000	0.00007
	0.02000	0.02737	0.33000	0.02675	2.50000	0.02671

Passive power(pJ) for c2p rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	(!c2p_en * pad * p2c)	0.02000	-0.00009	0.33000	-0.00006	2.50000	-0.00007
	(!c2p_en * pad * p2c)	0.02000	-0.02251	0.33000	-0.02360	2.50000	-0.02387
	(!c2p_en * !pad * !p2c)	0.02000	-0.00009	0.33000	-0.00007	2.50000	-0.00007
	(!c2p_en * !pad * !p2c)	0.02000	-0.02251	0.33000	-0.02360	2.50000	-0.02387
sg13g2_IOPadInOut30mA	(!c2p_en * pad * p2c)	0.02000	-0.00010	0.33000	-0.00006	2.50000	-0.00007
	(!c2p_en * pad * p2c)	0.02000	-0.02251	0.33000	-0.02361	2.50000	-0.02387
	(!c2p_en * !pad * !p2c)	0.02000	-0.00010	0.33000	-0.00007	2.50000	-0.00007
	(!c2p_en * !pad * !p2c)	0.02000	-0.02251	0.33000	-0.02361	2.50000	-0.02387
sg13g2_IOPadInOut4mA	(!c2p_en * pad * p2c)	0.02000	-0.00009	0.33000	-0.00007	2.50000	-0.00007
	(!c2p_en * pad * p2c)	0.02000	-0.02244	0.33000	-0.02360	2.50000	-0.02387
	(!c2p_en * !pad * !p2c)	0.02000	-0.00009	0.33000	-0.00007	2.50000	-0.00007
	(!c2p_en * !pad * !p2c)	0.02000	-0.02243	0.33000	-0.02360	2.50000	-0.02387

Passive power(pJ) for c2p falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	(!c2p_en * pad * p2c)	0.02000	0.00009	0.33000	0.00006	2.50000	0.00007
	(!c2p_en * pad * p2c)	0.02000	0.02737	0.33000	0.02676	2.50000	0.02671
	(!c2p_en * !pad * !p2c)	0.02000	0.00009	0.33000	0.00007	2.50000	0.00007
	(!c2p_en * !pad * !p2c)	0.02000	0.02737	0.33000	0.02676	2.50000	0.02671
sg13g2_IOPadInOut30mA	(!c2p_en * pad * p2c)	0.02000	0.00010	0.33000	0.00006	2.50000	0.00007
	(!c2p_en * pad * p2c)	0.02000	0.02737	0.33000	0.02676	2.50000	0.02671
	(!c2p_en * !pad * !p2c)	0.02000	0.00010	0.33000	0.00007	2.50000	0.00007
	(!c2p_en * !pad * !p2c)	0.02000	0.02737	0.33000	0.02676	2.50000	0.02671
sg13g2_IOPadInOut4mA	(!c2p_en * pad * p2c)	0.02000	0.00009	0.33000	0.00007	2.50000	0.00007
	(!c2p_en * pad * p2c)	0.02000	0.02737	0.33000	0.02675	2.50000	0.02671
	(!c2p_en * !pad * !p2c)	0.02000	0.00009	0.33000	0.00007	2.50000	0.00007
	(!c2p_en * !pad * !p2c)	0.02000	0.02737	0.33000	0.02675	2.50000	0.02671

Passive power(pJ) for c2p_en rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	1.25828	0.33000	1.25842	2.50000	1.26501
	0.02000	0.04451	0.33000	0.04882	2.50000	0.13154
sg13g2_IOPadInOut30mA	0.02000	2.00700	0.33000	2.00972	2.50000	2.01192
	0.02000	0.04451	0.33000	0.04882	2.50000	0.13147
sg13g2_IOPadInOut4mA	0.02000	0.61973	0.33000	0.61953	2.50000	0.62011
	0.02000	0.04450	0.33000	0.04881	2.50000	0.13151

Passive power(pJ) for c2p_en falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	0.27636	0.33000	0.27557	2.50000	0.27317
	0.02000	0.10276	0.33000	0.10823	2.50000	0.19105
sg13g2_IOPadInOut30mA	0.02000	0.26954	0.33000	0.26886	2.50000	0.26617
	0.02000	0.10277	0.33000	0.10824	2.50000	0.19113
sg13g2_IOPadInOut4mA	0.02000	0.29381	0.33000	0.29287	2.50000	0.29056
	0.02000	0.10278	0.33000	0.10822	2.50000	0.19097

Passive power(pJ) for pad rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	5.27597	0.33000	5.15565	2.50000	4.92411
	0.02000	1.89301	0.33000	16.28530	2.50000	30.15810
	0.02000	4.86827	0.33000	5.54010	2.50000	7.22589
	0.02000	1.85601	0.33000	16.26200	2.50000	30.17550
sg13g2_IOPadInOut30mA	0.02000	8.02507	0.33000	7.54518	2.50000	6.90732
	0.02000	3.69575	0.33000	31.51310	2.50000	14.23860
	0.02000	7.24610	0.33000	8.11320	2.50000	8.24592
	0.02000	3.65964	0.33000	31.49240	2.50000	14.22560
sg13g2_IOPadInOut4mA	0.02000	3.07455	0.33000	3.30633	2.50000	3.74832
	0.02000	1.89734	0.33000	9.10611	2.50000	21.88960
	0.02000	2.85163	0.33000	3.60699	2.50000	5.39874
	0.02000	1.86123	0.33000	9.07240	2.50000	21.99020
sg13g2_IOPadOut16mA	0.02000	4.94358	0.33000	4.80195	2.50000	4.64931
	0.02000	-0.02016	0.33000	-0.01002	2.50000	0.13424
sg13g2_IOPadOut30mA	0.02000	7.78521	0.33000	7.61256	2.50000	7.03950
	0.02000	-0.02016	0.33000	-0.01002	2.50000	0.13310
sg13g2_IOPadOut4mA	0.02000	2.60912	0.33000	2.51301	2.50000	2.48820
	0.02000	-0.02016	0.33000	-0.00962	2.50000	0.13335

Passive power(pJ) for pad falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadInOut16mA	0.02000	11.24310	0.33000	7.55593	2.50000	6.51853
	0.02000	0.22769	0.33000	0.24371	2.50000	0.39970
	0.02000	2.91114	0.33000	3.09096	2.50000	3.38177
	0.02000	0.12161	0.33000	0.13880	2.50000	0.23569
sg13g2_IOPadInOut30mA	0.02000	40.03380	0.33000	24.46550	2.50000	18.40310
	0.02000	0.22952	0.33000	0.24444	2.50000	0.38988
	0.02000	4.39915	0.33000	4.59171	2.50000	4.89371
	0.02000	0.12236	0.33000	0.13900	2.50000	0.22573
sg13g2_IOPadInOut4mA	0.02000	2.25819	0.33000	2.66728	2.50000	3.41698
	0.02000	0.23268	0.33000	0.25912	2.50000	0.42720
	0.02000	1.74782	0.33000	2.16602	2.50000	2.91195
	0.02000	0.12766	0.33000	0.15303	2.50000	0.26103
sg13g2_IOPadOut16mA	0.02000	11.72830	0.33000	7.29715	2.50000	5.96180
	0.02000	0.07593	0.33000	0.08982	2.50000	0.23376
sg13g2_IOPadOut30mA	0.02000	41.36780	0.33000	24.59110	2.50000	18.28580
	0.02000	0.07592	0.33000	0.08982	2.50000	0.23258
sg13g2_IOPadOut4mA	0.02000	1.92630	0.33000	1.87268	2.50000	1.85699
	0.02000	0.07596	0.33000	0.08984	2.50000	0.23179

INPUT



*sg13g2_io_slow_1p35V_3p0V_125C Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C, Voltage 1.35, Temp 125.00*

Truth Table

INPUT	OUTPUT
pad	p2c
0	0
1	1

Footprint

Cell Name	Area
sg13g2_IOPadIn	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)	Max Cap(pf)
	pad	p2c
sg13g2_IOPadIn	0.23620	0.24000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadIn	369.32000	1262.72200	3110.82000

Delay Information

Delay(ns) to p2c rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad->p2c (RR)	0.12000	0.02400	0.29452	0.80000	0.14400	0.51731	5.00000	0.24000	0.92907

Delay(ns) to p2c falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad->p2c (FF)	0.12000	0.02400	0.30118	0.80000	0.14400	0.50794	5.00000	0.24000	0.84261

Power Information

Internal switching power(pJ) to p2c rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad	0.12000	0.02400	-0.53033	0.80000	0.14400	-0.48511	5.00000	0.24000	-0.10551
	pad	0.12000	0.02400	0.03001	0.80000	0.14400	0.03101	5.00000	0.24000	0.03100

Internal switching power(pJ) to p2c falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadIn	pad	0.12000	0.02400	0.89869	0.80000	0.14400	0.94174	5.00000	0.24000	1.31991
	pad	0.12000	0.02400	0.07242	0.80000	0.14400	0.07684	5.00000	0.24000	0.09789

SG13G2_CORNER



sg13g2_io_slow_1p35V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C,
Voltage 1.35, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Corner	32400.00000

Pin Capacitance Information

SG13G2_FILLER10000



sg13g2_io_slow_1p35V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C,
Voltage 1.35, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Filler10000	18000.00000

Pin Capacitance Information

SG13G2_FILLER1000



sg13g2_io_slow_1p35V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C,
Voltage 1.35, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Filler1000	900.00000

Pin Capacitance Information

SG13G2_FILLER2000



sg13g2_io_slow_1p35V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C,
Voltage 1.35, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Filler2000	1800.00000

Pin Capacitance Information

SG13G2_FILLER200



sg13g2_io_slow_1p35V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C,
Voltage 1.35, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Filler200	180.00000

Pin Capacitance Information

SG13G2_FILLER4000



sg13g2_io_slow_1p35V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C,
Voltage 1.35, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Filler4000	3600.00000

Pin Capacitance Information

SG13G2_FILLER400



sg13g2_io_slow_1p35V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C,
Voltage 1.35, Temp 125.00

Footprint

Cell Name	Area
sg13g2_Filler400	360.00000

Pin Capacitance Information

SG13G2_IOPADANALOG



sg13g2_io_slow_1p35V_3p0V_125C

Cell Library: Process

sg13g2_io_slow_1p35V_3p0V_125C,

Voltage 1.35, Temp 125.00

Truth Table

INPUT		OUTPUT	
pad	padres	pad	padres
x	x	-	-

Footprint

Cell Name	Area
sg13g2_IOPadAnalog	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)	
	pad	padres	pad	padres
sg13g2_IOPadAnalog	0.78661	0.59885	30000.00000	30000.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadAnalog	0.00000	5820.60000	11641.20000

Delay Information

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadAnalog	pad->pad (R)	10.00000	500.00000	1000.00000	200.00000	30000.00000	1000.00000	200.00000	30000.00000	1000.00000

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadAnalog	pad->pad (F)	10.00000	500.00000	1000.00000	200.00000	30000.00000	1000.00000	200.00000	30000.00000	1000.00000

SG13G2_IOPADIOVDD



sg13g2_io_slow_1p35V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C,
Voltage 1.35, Temp 125.00

Footprint

Cell Name	Area
sg13g2_IOPadIOVdd	14400.00000

Pin Capacitance Information

SG13G2_IOPADIOVSS



sg13g2_io_slow_1p35V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C,
Voltage 1.35, Temp 125.00

Footprint

Cell Name	Area
sg13g2_IOPadIOVss	14400.00000

Pin Capacitance Information

SG13G2_IOPADVDD



sg13g2_io_slow_1p35V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C,
Voltage 1.35, Temp 125.00

Footprint

Cell Name	Area
sg13g2_IOPadVdd	14400.00000

Pin Capacitance Information

SG13G2_IOPADVSS



sg13g2_io_slow_1p35V_3p0V_125C
Cell Library: Process
sg13g2_io_slow_1p35V_3p0V_125C,
Voltage 1.35, Temp 125.00

Footprint

Cell Name	Area
sg13g2_IOPadVss	14400.00000

Pin Capacitance Information

TRI_OUTx



*sg13g2_io_slow_1p35V_3p0V_125C Cell Library:
Process sg13g2_io_slow_1p35V_3p0V_125C, Voltage
1.35, Temp 125.00*

Truth Table

INPUT		OUTPUT
c2p	c2p_en	pad
-	0	HiZ
0	1	0
1	1	1

Footprint

Cell Name	Area
sg13g2_IOPadTriOut16mA	14400.00000
sg13g2_IOPadTriOut30mA	14400.00000
sg13g2_IOPadTriOut4mA	14400.00000

Pin Capacitance Information

Cell Name	Pin Cap(pf)		Max Cap(pf)
	c2p	c2p_en	pad
sg13g2_IOPadTriOut16mA	0.02662	0.02609	20.00000
sg13g2_IOPadTriOut30mA	0.02662	0.02609	40.00000
sg13g2_IOPadTriOut4mA	0.02662	0.02610	12.00000

Leakage Information

Cell Name	Leakage(pW)		
	Min.	Avg	Max.
sg13g2_IOPadTriOut16mA	918.88900	7106.40740	14973.90000
sg13g2_IOPadTriOut30mA	1325.67000	7773.88500	14975.10000
sg13g2_IOPadTriOut4mA	564.11500	6537.73970	14974.20000

Delay Information

Delay(ns) to pad rising :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p->pad (RR)	0.02000	1.00000	2.24455	0.33000	9.00000	3.77601	2.50000	20.00000	5.65290
	c2p_en->pad (FR)	0.02000	1.00000	1.61042	0.33000	9.00000	1.67378	2.50000	20.00000	1.85835
	c2p_en->pad (RR)	0.02000	1.00000	2.23562	0.33000	9.00000	3.80172	2.50000	20.00000	5.73632
sg13g2_IOPadTriOut30mA	c2p->pad (RR)	0.02000	2.00000	2.73791	0.33000	18.00000	4.58046	2.50000	40.00000	6.61065
	c2p_en->pad (FR)	0.02000	2.00000	1.86514	0.33000	18.00000	1.92963	2.50000	40.00000	2.11691
	c2p_en->pad (RR)	0.02000	2.00000	2.71503	0.33000	18.00000	4.59062	2.50000	40.00000	6.65371
sg13g2_IOPadTriOut4mA	c2p->pad (RR)	0.02000	1.00000	2.30504	0.33000	5.00000	5.01224	2.50000	12.00000	9.70533
	c2p_en->pad (FR)	0.02000	1.00000	1.34630	0.33000	5.00000	1.41165	2.50000	12.00000	1.59599
	c2p_en->pad (RR)	0.02000	1.00000	2.31664	0.33000	5.00000	5.10127	2.50000	12.00000	9.93797

Delay(ns) to pad falling :

Cell Name	Timing Arc(Dir)	Delay(ns)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p->pad (FF)	0.02000	1.00000	1.66432	0.33000	9.00000	3.04615	2.50000	20.00000	5.13128
	c2p_en->pad (FF)	0.02000	1.00000	1.61527	0.33000	9.00000	1.70499	2.50000	20.00000	1.95780
	c2p_en->pad (RF)	0.02000	1.00000	1.41563	0.33000	9.00000	2.90202	2.50000	20.00000	4.77784
sg13g2_IOPadTriOut30mA	c2p->pad (FF)	0.02000	2.00000	2.21822	0.33000	18.00000	3.76554	2.50000	40.00000	5.91111
	c2p_en->pad (FF)	0.02000	2.00000	2.49537	0.33000	18.00000	2.57416	2.50000	40.00000	2.83138
	c2p_en->pad (RF)	0.02000	2.00000	1.68446	0.33000	18.00000	3.38636	2.50000	40.00000	5.38758
sg13g2_IOPadTriOut4mA	c2p->pad (FF)	0.02000	1.00000	1.77517	0.33000	5.00000	4.48222	2.50000	12.00000	9.41270
	c2p_en->pad (FF)	0.02000	1.00000	0.88667	0.33000	5.00000	0.96841	2.50000	12.00000	1.24205
	c2p_en->pad (RF)	0.02000	1.00000	1.77488	0.33000	5.00000	4.54671	2.50000	12.00000	9.34692

Power Information

Internal switching power(pJ) to pad rising :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p	0.02000	1.00000	4.92147	0.33000	9.00000	4.81224	2.50000	20.00000	4.90749
	c2p	0.02000	1.00000	0.06462	0.33000	9.00000	0.06905	2.50000	20.00000	0.21145
	c2p_en	0.02000	1.00000	4.46721	0.33000	9.00000	5.10630	2.50000	20.00000	6.00051
	c2p_en	0.02000	1.00000	0.02839	0.33000	9.00000	0.03767	2.50000	20.00000	0.19690
sg13g2_IOPadTriOut30mA	c2p	0.02000	2.00000	7.75828	0.33000	18.00000	7.59792	2.50000	40.00000	7.39110
	c2p	0.02000	2.00000	0.06462	0.33000	18.00000	0.06905	2.50000	40.00000	0.21148
	c2p_en	0.02000	2.00000	6.89211	0.33000	18.00000	7.39572	2.50000	40.00000	8.01000
	c2p_en	0.02000	2.00000	0.02840	0.33000	18.00000	0.03767	2.50000	40.00000	0.19718
sg13g2_IOPadTriOut4mA	c2p	0.02000	1.00000	2.60473	0.33000	5.00000	2.50872	2.50000	12.00000	2.38566
	c2p	0.02000	1.00000	0.06466	0.33000	5.00000	0.06911	2.50000	12.00000	0.21157
	c2p_en	0.02000	1.00000	2.37150	0.33000	5.00000	2.81283	2.50000	12.00000	3.60243
	c2p_en	0.02000	1.00000	0.02844	0.33000	5.00000	0.03771	2.50000	12.00000	0.19831

Internal switching power(pJ) to pad falling :

Cell Name	Input	Power(pJ)								
		Slew(ns)	Load(pf)	First	Slew(ns)	Load(pf)	Mid	Slew(ns)	Load(pf)	Last
sg13g2_IOPadTriOut16mA	c2p	0.02000	1.00000	10.91290	0.33000	9.00000	7.04883	2.50000	20.00000	5.74144
	c2p	0.02000	1.00000	0.15015	0.33000	9.00000	0.15508	2.50000	20.00000	0.29853
	c2p_en	0.02000	1.00000	2.60000	0.33000	9.00000	2.59940	2.50000	20.00000	2.59858
	c2p_en	0.02000	1.00000	0.04524	0.33000	9.00000	0.04955	2.50000	20.00000	0.13397
sg13g2_IOPadTriOut30mA	c2p	0.02000	2.00000	39.57620	0.33000	18.00000	23.90530	2.50000	40.00000	17.54800
	c2p	0.02000	2.00000	0.15017	0.33000	18.00000	0.15510	2.50000	40.00000	0.29855
	c2p_en	0.02000	2.00000	4.07644	0.33000	18.00000	4.07986	2.50000	40.00000	4.07774
	c2p_en	0.02000	2.00000	0.04526	0.33000	18.00000	0.04957	2.50000	40.00000	0.13383
sg13g2_IOPadTriOut4mA	c2p	0.02000	1.00000	1.88736	0.33000	5.00000	1.87921	2.50000	12.00000	1.88042
	c2p	0.02000	1.00000	0.15036	0.33000	5.00000	0.15529	2.50000	12.00000	0.29850
	c2p_en	0.02000	1.00000	1.36503	0.33000	5.00000	1.36348	2.50000	12.00000	1.36367
	c2p_en	0.02000	1.00000	0.04544	0.33000	5.00000	0.04977	2.50000	12.00000	0.13307

Passive power(pJ) for c2p rising :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	-0.02253	0.33000	-0.02384	2.50000	-0.02418
sg13g2_IOPadTriOut30mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	-0.02253	0.33000	-0.02384	2.50000	-0.02418
sg13g2_IOPadTriOut4mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	-0.02253	0.33000	-0.02384	2.50000	-0.02418

Passive power(pJ) for c2p falling :

Cell Name	Power(pJ)					
	Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	0.02719	0.33000	0.02653	2.50000	0.02655
sg13g2_IOPadTriOut30mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	0.02717	0.33000	0.02653	2.50000	0.02652
sg13g2_IOPadTriOut4mA	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	0.02000	0.02716	0.33000	0.02654	2.50000	0.02621

Passive power(pJ) for c2p rising (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	-0.02253	0.33000	-0.02384	2.50000	-0.02418
sg13g2_IOPadTriOut30mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	-0.02253	0.33000	-0.02384	2.50000	-0.02418
sg13g2_IOPadTriOut4mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	-0.02253	0.33000	-0.02384	2.50000	-0.02418

Passive power(pJ) for c2p falling (conditional):

Cell Name	When	Power(pJ)					
		Slew(ns)	First	Slew(ns)	Mid	Slew(ns)	Last
sg13g2_IOPadTriOut16mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	0.02719	0.33000	0.02653	2.50000	0.02655
sg13g2_IOPadTriOut30mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	0.02717	0.33000	0.02653	2.50000	0.02652
sg13g2_IOPadTriOut4mA	!c2p_en	0.02000	0.00000	0.33000	0.00000	2.50000	0.00000
	!c2p_en	0.02000	0.02716	0.33000	0.02654	2.50000	0.02621